Week 03 teacher guide

Rates and percents

PowerPoint slides 202-297 | Student workbook pp. 25-35

A unit rate describes an amount per one unit. Percent means per hundred. Label the whole before deciding whether to find a part, a whole, or a percent.

The seven days

Day 015 · pp. 25-27 · Saturday · 20 questions,including 2 challenges
Rates and unit rates

Day 016 · pp. 28-30 · Sunday · 20 questions,including 2 challenges
Percent as a rate per hundred

Day 017 · p. 31 · Monday · 20 questions,including 2 challenges
Rates and percents

Day 018 · p. 32 · Tuesday · 20 questions,including 2 challenges
Rates and percents

Day 019 · p. 33 · Wednesday · 25 questions,including 2 challenges
Rates and percents

Day 020 · p. 34 · Thursday · 20 questions,including 2 challenges
Rates and percents

Day 021 · p. 35 · Friday · 20 questions,including 2 challenges
Weekly check

Use one day at a time

Each weekend lesson uses a three-page packet; each weekday uses one question page. Work on separate plain paper, labeling the Day number and item IDs. Weekend lessons and ordinary weekdays contain 20 questions including C1 and C2; Wednesday contains 25. Allow 30-45 minutes on ordinary weekdays and 40-55 minutes on Wednesday. Split a demanding day into two sittings if needed.

The weekend packet includes models and worked examples, or a scheduled assessment alternative. Read a worked example after an attempt. Keep answer-review slides out of view until the corresponding work is complete. Weekly answer review is optional after Friday and does not add to the 80-minute weekend lesson.

Built-in review

Day 017: Ratios and rates (9, 10, 11, 12)

Day 018: Percent practice (9, 10, 11, 12)

Day 019: Weekly puzzle (13)

Day 020: Decimal practice (9, 10, 11, 12)

On Friday, record routine accuracy out of 18 separately from the two challenge explanations. For each challenge, note whether the learner connected the steps, used a valid model or calculation, and justified the conclusion. Revisit a missed model, then check with a fresh item later.

Assessment alternatives: Mini Check A (p. 29). Replace ordinary practice or use a separate assessment sitting; do not add these on top of the assigned day.

Saturday. Rates and unit rates

Day 015 · pp. 25-27 | Slides 202-230

Goal: I can find an amount per one unit and use it to compare rates.

Check first: Equivalent ratios and decimal division by whole numbers.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 202-203 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 204-205 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 206-207 · Explore · P1, P2, P3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 208 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 209-210 · Explore · P4, P5, P6, P7

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 211-213 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 214 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 215 · Compare methods

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 216 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 217 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 218-220 · Guided practice · X1, X2, X3

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 221-222 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 223-226 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 227 · Summary · S

Ask for one sentence or annotated example in the learner’s own words. Listen for the main relationship.

Slides 228 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 229 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 230 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Saturday. Discussion and support

Investigation T: Two unit rates, one relationship

Watch and discuss; this is a projected investigation, not an extra assigned question.

A printer makes 12 cards in 5 minutes. Find cards per minute and minutes per card. Use each rate to answer a different question: cards in 20 minutes, and time for 30 cards.

1. Name each division and its units.
2. Find both unit rates.
3. Use cards per minute to convert time to cards.
4. Use minutes per card to convert cards to time.

Ask: Which units cancel in each multiplication? Why are the two unit rates reciprocals?

If stuck: Write one rate with cards on top, and the other with minutes on top.

Expected: $\frac{12}{5}$ cards/min and $\frac{5}{12}$ min/card; 48 cards in 20 minutes; $\frac{25}{2}$ minutes for 30 cards.

C1 · Challenge discussion

Ask: Why is a larger distance not enough to identify the faster runner?

If stuck: Compare both distances for 10 minutes first.

Expected: Same speed: 12 miles per hour; each travels 15 miles in 75 minutes

C2 · Challenge discussion

Ask: Can the higher-priced pack still have the lower unit price?

If stuck: Use twelve notebooks as a common quantity to check the unit-price comparison.

Expected: A:$1.50 each; B:$1.40 each; B saves $1.20 on 12

What to notice

Labels dollars per apple. → Ask what one in the rate refers to.

Uses scaling or divides to one. → Accept either, then connect them in a table.

Common error: Divides in the reverse direction. Repair: Read the desired rate aloud before dividing.

Sunday. Percent as a rate per hundred

Day 016 · pp. 28-30 | Slides 231-261

Goal: I can find a percent, a part, or a whole using matching amounts.

Check first: Equivalent ratios and unit rates.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 231-232 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 233-234 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 235-236 · Explore · P1, P2, P3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 237 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 238-239 · Explore · P4, P5, P6

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 240-242 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 243 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 244 · Compare methods

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 245 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 246 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 247-250 · Guided practice · X1, X2, X3, X4

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 251-252 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 253-256 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 257 · Summary · S

Ask for one sentence or annotated example in the learner’s own words. Listen for the main relationship.

Slides 258 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 259 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 260 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Slides 261 · Assessment alternative

Replace ordinary practice with this check, or use a separate sitting. It is not extra assigned work.

Sunday. Discussion and support

Investigation T: One percent relationship, three unknowns

Watch and discuss; this is a projected investigation, not an extra assigned question.

Use one percent diagram to solve three linked questions: 20% of 75 is what? 15 is what percent of 75? 15 is 20% of what? Explain what stays the same.

1. Label 75 as 100% and find a 20% group.
2. One group gives the part 15.
3. Use 15 out of 75 to find 20%.
4. Reverse the grouping to recover the whole 75.

Ask: Which number names 100% in every version? What would go wrong if we treated 24 as the whole?

If stuck: Draw the same five equal boxes for all three questions; only hide a different label each time.

Expected: 15; 20%; 75. The part, whole, and percent relationship is the same; only the unknown changes.

C1 · Challenge discussion

Ask: Which amount represents 100% in both parts of the problem?

If stuck: Find 10% first, then the whole.

Expected: 80 books total; 20 science books; fiction has 4 more

C2 · Challenge discussion

Ask: What is the whole for each percent?

If stuck: Separate each percent into 100% and the extra part.

Expected: Both collect 60 cans

What to notice

Labels the given amount as part or whole. → Ask which amount belongs at 100%.

Combines benchmark percentages. → Accept a unit-percent method or a benchmark method.

Common error: Multiplies the known part by the percent. Repair: Place the known part and its percent on aligned lines.

Slide-order index

Follow this numerical order. The session plans explain how to teach each phase; full scripts remain in PowerPoint speaker notes. Item IDs match the student workbook.

Saturday

202. Week 3, Saturday
Watch and discuss

203. Today
Watch and discuss

204. Recall
Day 015 · Workbook p. 25 · R1, R2, R3, R4

205. Recall: answers
Day 015 · Workbook p. 25 · R1, R2, R3, R4

206. Whole-number unit rate
Day 015 · Workbook p. 25 · P1

207. Decimal unit rate
Day 015 · Workbook p. 25 · P2, P3

208. The model
Watch and discuss

209. Both unit-rate directions
Day 015 · Workbook pp. 25, 26 · P4, P5

210. Time below one hour
Day 015 · Workbook p. 26 · P6, P7

211. Worked example · 1 of 3
Watch and discuss · worked example

212. Worked example · 2 of 3
Watch and discuss · worked example

213. Worked example · 3 of 3
Watch and discuss · worked example

214. Your turn
Day 015 · Workbook p. 26 · F

215. Two ways to see it
Watch and discuss · compare the same example

216. A claim to check
Day 015 · Workbook p. 26 · D

217. Words and meaning
Watch and discuss

218. Guided practice · X1
Day 015 · Workbook p. 26 · X1

219. Guided practice · X2
Day 015 · Workbook p. 26 · X2

220. Guided practice · X3
Day 015 · Workbook p. 26 · X3

221. Two unit rates, one relationship
Watch and discuss · draw on separate paper

222. Two unit rates, one relationship
Watch and discuss · draw on separate paper

223. Challenge C1
Day 015 · Workbook p. 27 · C1

224. Challenge C1 · Discuss
Day 015 · Workbook p. 27 · C1

225. Challenge C2
Day 015 · Workbook p. 27 · C2

226. Challenge C2 · Discuss
Day 015 · Workbook p. 27 · C2

227. In your own words
Day 015 · Workbook p. 27 · S

228. Exit ticket
Day 015 · Workbook p. 27 · E

229. The week ahead
Days 017-021 · Workbook pp. 31-35

230. Tutor reference
Tutor only;hidden in slide show

Sunday

231. Week 3, Sunday
Watch and discuss

232. Today
Watch and discuss

233. Recall
Day 016 · Workbook p. 28 · R1, R2, R3, R4

234. Recall: answers
Day 016 · Workbook p. 28 · R1, R2, R3, R4

235. Benchmark percents of a whole
Day 016 · Workbook p. 28 · P1

236. Find percent from part and whole
Day 016 · Workbook p. 28 · P2, P3

237. The model
Watch and discuss

238. Percent above one hundred
Day 016 · Workbook p. 28 · P4, P5

239. Decimal result
Day 016 · Workbook p. 28 · P6

240. Worked example · 1 of 3
Watch and discuss · worked example

241. Worked example · 2 of 3
Watch and discuss · worked example

242. Worked example · 3 of 3
Watch and discuss · worked example

243. Your turn
Day 016 · Workbook p. 28 · F

244. Two ways to see it
Watch and discuss · compare the same example

245. A claim to check
Day 016 · Workbook p. 28 · D

246. Words and meaning
Watch and discuss

247. Guided practice · X1
Day 016 · Workbook p. 28 · X1

248. Guided practice · X2
Day 016 · Workbook p. 28 · X2

249. Guided practice · X3
Day 016 · Workbook p. 28 · X3

250. Guided practice · X4
Day 016 · Workbook p. 28 · X4

251. One percent relationship, three unknowns
Watch and discuss · draw on separate paper

252. One percent relationship, three unknowns
Watch and discuss · draw on separate paper

253. Challenge C1
Day 016 · Workbook p. 30 · C1

254. Challenge C1 · Discuss
Day 016 · Workbook p. 30 · C1

255. Challenge C2
Day 016 · Workbook p. 30 · C2

256. Challenge C2 · Discuss
Day 016 · Workbook p. 30 · C2

257. In your own words
Day 016 · Workbook p. 28 · S

258. Exit ticket
Day 016 · Workbook p. 28 · E

259. The week ahead
Days 017-021 · Workbook pp. 31-35

260. Tutor reference
Tutor only;hidden in slide show

261. Optional assessment · Mini Check A
Day 016 · Workbook pp. 29-29 · Mini Check A

Answer review

262. Week 3 answers
Next lesson: slide 298. Use this review section after Friday.

263. Rates and unit rates · 1 of 5
Day 015 · Workbook p. 25 · Items R1, R2, R3, R4

264. Rates and unit rates · 2 of 5
Day 015 · Workbook p. 25 · Items P1, P2, P3, P4

265. Rates and unit rates · 3 of 5
Day 015 · Workbook p. 26 · Items P5, P6, P7, F, D

266. Rates and unit rates · 4 of 5
Day 015 · Workbook pp. 26, 27 · Items X1, X2, X3, S

267. Rates and unit rates · 5 of 5
Day 015 · Workbook p. 27 · Items E, C1, C2

268. Percent as a rate per hundred · 1 of 5
Day 016 · Workbook p. 28 · Items R1, R2, R3, R4

269. Percent as a rate per hundred · 2 of 5
Day 016 · Workbook p. 28 · Items P1, P2, P3, P4, P5

270. Percent as a rate per hundred · 3 of 5
Day 016 · Workbook p. 28 · Items P6, F, D, X1, X2

271. Percent as a rate per hundred · 4 of 5
Day 016 · Workbook p. 28 · Items X3, X4, S, E

272. Percent as a rate per hundred · 5 of 5
Day 016 · Workbook p. 30 · Items C1, C2

273. Mini Check A · 1 of 2
Day 016 · Workbook p. 29 · Items 1, 2, 3, 4

274. Mini Check A · 2 of 2
Day 016 · Workbook p. 29 · Items 5, 6, 7, 8

275. Rates and percents · 1 of 5
Day 017 · Workbook p. 31 · Items 1, 2, 3, 4, 5

276. Rates and percents · 2 of 5
Day 017 · Workbook p. 31 · Items 6, 7, 8, 9, 10

277. Rates and percents · 3 of 5
Day 017 · Workbook p. 31 · Items 11, 12, 13, 14

278. Rates and percents · 4 of 5
Day 017 · Workbook p. 31 · Items 15, 16, 17, 18

279. Rates and percents · 5 of 5
Day 017 · Workbook p. 31 · Items C1, C2

280. Rates and percents · 1 of 4
Day 018 · Workbook p. 32 · Items 1, 2, 3, 4, 5

281. Rates and percents · 2 of 4
Day 018 · Workbook p. 32 · Items 6, 7, 8, 9, 10

282. Rates and percents · 3 of 4
Day 018 · Workbook p. 32 · Items 11, 12, 13, 14, 15

283. Rates and percents · 4 of 4
Day 018 · Workbook p. 32 · Items 16, 17, 18, C1, C2

284. Rates and percents · 1 of 5
Day 019 · Workbook p. 33 · Items 1, 2, 3, 4, 5

285. Rates and percents · 2 of 5
Day 019 · Workbook p. 33 · Items 6, 7, 8, 9, 10

286. Rates and percents · 3 of 5
Day 019 · Workbook p. 33 · Items 11, 12, 13, 14, 15

287. Rates and percents · 4 of 5
Day 019 · Workbook p. 33 · Items 16, 17, 18, 19, 20

288. Rates and percents · 5 of 5
Day 019 · Workbook p. 33 · Items 21, 22, 23, C1, C2

289. Rates and percents · 1 of 4
Day 020 · Workbook p. 34 · Items 1, 2, 3, 4, 5

290. Rates and percents · 2 of 4
Day 020 · Workbook p. 34 · Items 6, 7, 8, 9, 10

291. Rates and percents · 3 of 4
Day 020 · Workbook p. 34 · Items 11, 12, 13, 14, 15

292. Rates and percents · 4 of 4
Day 020 · Workbook p. 34 · Items 16, 17, 18, C1, C2

293. Weekly check · 1 of 5
Day 021 · Workbook p. 35 · Items 1, 2, 3, 4, 5

294. Weekly check · 2 of 5
Day 021 · Workbook p. 35 · Items 6, 7, 8, 9

295. Weekly check · 3 of 5
Day 021 · Workbook p. 35 · Items 10, 11, 12, 13

296. Weekly check · 4 of 5
Day 021 · Workbook p. 35 · Items 14, 15, 16, 17, 18

297. Weekly check · 5 of 5
Day 021 · Workbook p. 35 · Items C1, C2

Complete weekly answers

Match Day,item ID,and page before checking. Equivalent correct forms and valid alternative methods are welcome. Computational keys are compact; challenges and conceptual items include reasoning. No challenge is complete with an unsupported answer alone.

Day 015 · Rates and unit rates

Workbook pp. 25-27 · assigned set

R1 · F-ADD
$\frac{11}{12}$

R2 · R-WHOLE
8 yellow; 20 purple

R3 · F-DIV
$\frac{1}{6}$ liter

R4 · R-EQUIV
3 cups

P1 · R-RATE
5 kilometers per hour

P2 · R-RATE
\$0.75 per apple

P3 · R-RATE
$\frac{2}{5}$ cup per serving

P4 · R-RATE
5 cards per minute; $\frac{1}{5}$ minute per card

P5 · R-COMP
Pack B: \$0.35 per pencil

Reason: A costs \$0.40 each. B costs \$0.35 each, saving \$0.05 per pencil.

P6 · R-RATE
9 miles per hour

P7 · R-TABLE
0, 8, and 20 meters

F · R-RATE
2; 10 miles per hour

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · R-RATE
No. \$6 ÷ 8 apples = \$0.75 per apple. $\frac{4}{3}$ describes apples per dollar.

Reason: She reverses the requested quantities. Write dollars above the division and apples below it.

X1 · R-RATE
10 km per hour

X2 · R-RATE
\$0.35 per marker

X3 · R-RATE
7 cards/min; $\frac{1}{7}$ min/card

S · EXPLAIN
Answers vary. A unit rate names how much of one quantity goes with one unit of another. Its units show the division order.

Reason: Accept an accurate explanation supported by a matching example.

E · R-RATE
12 miles per hour; 5 minutes per mile

Reason: Nine miles in three 15-minute groups means 3 miles per 15 minutes. Four groups give 12 miles in an hour.

C1 · R-RATE
Same speed: 12 miles per hour; each travels 15 miles in 75 minutes

Reason: A covers 2 miles per 10 minutes; B does too. One hour has six 10-minute intervals, giving 12 miles. Seventy-five minutes is $\frac{5}{4}$ hour, so $12 \times \frac{5}{4} = 15$ miles.

C2 · R-RATE
A:\$1.50 each; B:\$1.40 each; B saves \$1.20 on 12

Reason: Divide each pack price by its count. Twelve notebooks cost 3 × \$6 = \$18 at A and 2 × \$8.40 = \$16.80 at B. The difference is \$1.20.

Day 016 · Percent as a rate per hundred

Workbook pp. 28-30 · assigned set

R1 · F-DIV
$\frac{1}{18}$; stories vary

R2 · R-EQUIV
24 runners

R3 · F-LINE
At 1 $\frac{3}{4}$, the seventh quarter-step from 0

R4 · R-TABLE

32 laps

P1 · P-PART

8, 20, 40, and 60 counters

P2 · P-PERCENT

30%

P3 · P-WHOLE

60 stickers

P4 · P-PART

60 minutes

P5 · P-PART

21 tiles

P6 · P-PART

$2.70

F · P-WHOLE

6 coins; 60 coins

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · P-WHOLE

The unknown whole is 60. Check: 40% of 60 is 24.

Reason: He treats the known part as the whole. Label 24 at 40% on a double number line and locate 100%.

X1 · P-PART

49; whole = 140

X2 · P-WHOLE

80

X3 · P-PERCENT

15%

X4 · P-PART

$1.30

S · EXPLAIN

Answers vary. Percent compares an amount with a whole split into 100 equal parts.

Reason: Accept an accurate explanation supported by a matching example.

E · P-WHOLE

60 marbles

Reason: Seven 5% groups contain 21 marbles. Each 5% group has 3. Twenty groups have 60.

C1 · P-WHOLE

80 books total; 20 science books; fiction has 4 more

Reason: Three 10% groups contain 24 books, so 10% = 8 and 100% = 80. A quarter of 80 is 20. Compare 24 fiction with 20 science.

C2 · P-PART

Both collect 60 cans

Reason: 125% is one whole plus one fourth: 48 + 12 = 60. 150% is one whole plus one half: 40 + 20 = 60. The percents act on different goals, so percent alone cannot decide the larger amount.

Day 016 · Mini Check A

Workbook p. 29 · assessment alternative

1 · F-COMP

$\frac{2}{9} < \frac{7}{9}$

Reason: With ninths fixed, compare two parts and seven parts.

2 · F-COMP

$\frac{7}{8}$

Reason: The gaps from 1 are one eighth and one fifth. One eighth is smaller.

3 · R-RATIO

5 : 2; 2 : 5

Reason: Keep the requested first color first.

4 · R-EQUIV

Yes

Reason: Multiply both numbers in 6 : 10 by one and one half.

5 · R-RATE

24 miles

6 · R-COMP

Pack B: $0.40 each

Reason: Compare forty-five cents with forty cents for one eraser.

7 · P-PART

42 points

8 · P-PART

28 tiles

Day 017 · Rates and percents

Workbook p. 31 · assigned set

1 · N-FACT

15

2 · N-FACT
35

3 · N-FACT
20

4 · N-FACT
45

5 · F-SCALE
Larger; 8

Reason: Four thirds takes one whole six plus one third of six.

6 · D-OPS
3.75

7 · N-FACT
36

8 · R-RATE
4 miles per hour

9 · R-RATE · review
24 miles

10 · R-COMP · review
Pack B: \$0.40 each

Reason: Compare forty-five cents with forty cents for one eraser.

11 · R-RATE · review
4 labels per minute; $\frac{1}{4}$ minute per label

12 · R-TABLE · review
0, 12, and 21 centimeters

13 · R-RATE
No. \$6 ÷ 8 apples = \$0.75 per apple. $\frac{4}{3}$ describes apples per dollar.

14 · R-RATE
8 km per hour

15 · R-RATE
\$0.65 per marker

16 · R-RATE
3 cards/min; $\frac{1}{3}$ min/card

17 · R-RATE
9 miles per hour

18 · R-RATE
Pack A: \$0.60 vs \$0.65

Reason: Divide each price by its count. Compare costs for one pencil, not pack prices.

C1 · R-RATE
$\frac{8}{3}$ labels/min; $\frac{3}{8}$ min/label; 15 minutes

Reason: The two unit rates reverse the division and units. Forty labels are five 8-label groups, so five 3-minute intervals take 15 minutes. $40 \times \frac{3}{8}$ checks the time.

C2 · R-RATE
6 extra miles; 15 miles altogether

Reason: Nine miles in 45 minutes is 3 miles per 15 minutes. Thirty more minutes is two such intervals, so 6 extra miles. Add 9 + 6 = 15 miles over 75 minutes.

Day 018 · Rates and percents

Workbook p. 32 · assigned set

1 · F-EQUIV
0.5; 50%

2 · F-EQUIV
0.25; 25%

3 · F-EQUIV
0.75; 75%

4 · F-EQUIV
0.125; 12.5%

5 · R-WHOLE
$\frac{3}{8}$

6 · G-VOL
36 cm³

7 · N-FACT
12

8 · P-PART
4, 20, and 30 counters

9 · P-PART · review
42 points

10 · P-PART · review
28 tiles

11 · P-WHOLE · review
40 shells

12 · P-PERCENT · review
75 %

13 · P-WHOLE
The unknown whole is 60. Check: 40% of 60 is 24.

14 · P-PART
30; whole = 100

15 · P-WHOLE
160

16 · P-PERCENT
15%

17 · P-PART
$1.35

18 · R-RATE
3 km per hour

C1 · P-PART
Reductions:$9 and $12. Both final prices are $36

Reason: Twenty percent of 45 is 9, leaving 36. One fourth of 48 is 12, also leaving 36. The discount amounts differ because the original prices differ.

C2 · P-PERCENT
37.5% bicycle; 25% walk; 37.5% neither

Reason: Twelve out of 32 is $\frac{3}{8}$ = 37.5%. Eight out of 32 is $\frac{1}{4}$ = 25%. Twenty students are in the two groups, leaving 12, or 37.5%, who do neither.

Day 019 · Rates and percents

Workbook p. 33 · assigned set

1 · N-SCALE
52

2 · N-SCALE
5

3 · N-SCALE
11

4 · N-SCALE
15

5 · R-RATE
4 miles per hour

6 · P-PERCENT
75 %

7 · R-TABLE
21

8 · F-LINE
Halfway

9 · F-ADD
$\frac{1}{2}$

10 · N-SCALE
70

11 · N-FACT
24

12 · G-VOL
72 cm³

13 · PUZZLE · review
40

14 · R-RATE
6 km per hour

15 · R-RATE
$0.55 per marker

16 · R-RATE
5 cards/min; $\frac{1}{5}$ min/card

17 · R-RATE
12 miles per hour

18 · R-RATE
Pack A: $0.55 vs $0.60

Reason: Divide each price by its count. Compare costs for one pencil, not pack prices.

19 · R-TABLE
0, 6, 12 meters

20 · P-PART
8; whole = 40

21 · P-WHOLE
80

22 · P-PERCENT
35%

23 · P-PART
$4.00

C1 · P-WHOLE
Whole 60 m²; beans 24 m²; paths 12 m²

Reason: If 40% = 24, then 10% = 6 and 100% = 60. After flowers, 36 m² remain. Beans take $\frac{2}{3} \times 36 = 24$ m², leaving 12 m² for paths. The three regions sum to 60.

C2 · R-RATE
A: 15 km/h; B: 18 km/h; B needs 40 minutes

Reason: A covers $10 \times \frac{60}{40} = 15$ km/h; B covers $9 \times \frac{60}{30} = 18$ km/h. Twelve km is $\frac{4}{3}$ of B's 9-km trip, so time is $30 \times \frac{4}{3} = 40$ minutes.

Day 020 · Rates and percents

Workbook p. 34 · assigned set

1 · N-EST
500

2 · N-EST
6

3 · N-EST
About 100

4 · N-EST
About 20

5 · R-RATE
4 miles per hour

6 · P-PERCENT
75 %

7 · P-WHOLE
40 shells

8 · F-DIV
6 pieces

9 · D-OPS · review
4.15

10 · D-OPS · review
1.70

11 · D-OPS · review
6.15

12 · D-OPS · review
2.70

13 · R-RATE
4 miles per hour

14 · P-WHOLE
The unknown whole is 60. Check: 40% of 60 is 24.

15 · R-RATE
5 km per hour

16 · R-RATE
$0.50 per marker

17 · R-RATE
6 cards/min; $\frac{1}{6}$ min/card

18 · R-RATE
42 miles per hour

C1 · P-PART
$3 reserved; 8 pens

Reason: Twenty percent of 15 is 3. The remaining budget is 12. Eight pens cost 8 × 1.50 = 12; a ninth would cost 13.50, which is too much.

C2 · P-WHOLE
42 liters needed; 7 minutes

Reason: If 30% = 18, then 10% = 6 and 100% = 60 liters. The missing amount is 60 − 18 = 42 liters. At 6 liters each minute, 42 ÷ 6 = 7 minutes.

Day 021 · Weekly check

Workbook p. 35 · assigned set

1 · R-RATE
4 miles per hour

2 · P-PERCENT
75 %

3 · P-WHOLE
40 shells

4 · R-RATE
4 miles per hour

5 · F-SCALE
Smaller; 6

Reason: Two thirds takes less than all nine.

6 · F-DIV
12 pieces

7 · R-RATIO
5 : 2; 2 : 5

Reason: Keep the requested first color first.

8 · F-ADD
$\frac{1}{3}$

9 · N-SCALE
7

10 · R-RATE
No. $6 ÷ 8 apples = $0.75 per apple. $\frac{4}{3}$ describes apples per dollar.

11 · R-RATE
4 km per hour

12 · R-RATE
$0.45 per marker

13 · R-RATE

7 cards/min; $\frac{1}{7}$ min/card

14 · R-RATE

28 miles per hour

15 · R-RATE

Pack A: \$0.55 vs \$0.60

Reason: Divide each price by its count. Compare costs for one pencil, not pack prices.

16 · R-TABLE

0, 30, 40 meters

17 · P-PART

96; whole = 160

18 · P-WHOLE

80

C1 · P-PART

30 students and 4 adults

Reason: 15% of 200 is 30 students. Thirty students make two 15-student groups, so two 2-adult groups require 4 adults. Total travelers would be 34.

C2 · P-PERCENT

The claim is correct; 60% was used

Reason: $\frac{12}{30} = \frac{2}{5} = 40\%$. The number used is $30 - 12 = 18$, and $\frac{18}{30} = \frac{3}{5} = 60\%$. Used and remaining percents sum to 100%.